

Markov analysis

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1 Markov Analysis

A fintech company provides historical data on its clients. Each client can be in one of the following states:

$$S = \{\text{active}, \text{inactive}, \text{default}\}.$$

The company wants to understand how its portfolio may evolve over time using the historical transitions observed in the data.

1. Estimate the transition matrix P using the historical data.
2. Determine the current portfolio vector π_0 , containing the number of clients in each state.
3. Suppose the company currently has 120 active clients, with no clients in the other states. Define the initial portfolio vector π_0 .
4. Using the estimated transition matrix, determine the expected number of clients in each state three months from now.

$$\pi_3 = \pi_0 P^3.$$

5. Based on your results, what can you tell the manager about the expected deterioration of the portfolio?

The database is available **here**.